FACT SHEET

Hormone Health NETWORK
Keep your body in balance

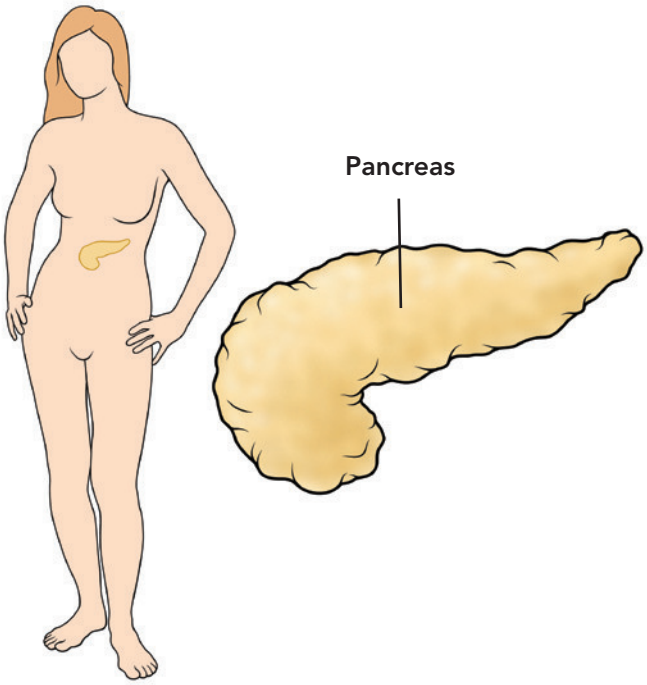
Diabetes and Low Blood Glucose (Hypoglycemia)

WHAT IS HYPOGLYCEMIA?

Hypoglycemia is the term for low blood glucose (sugar). Glucose is produced from the food you eat, and is the "fuel" that your brain and body need to function properly. Severe hypoglycemia can be dangerous and must be treated promptly.

DID YOU KNOW?

Hypoglycemia occurs most often in people who are taking medications to treat diabetes.



WHAT IS THE LINK BETWEEN DIABETES AND HYPOGLYCEMIA?

Diabetes is a disease in which blood glucose levels are too high. It is often treated with insulin—a hormone made by the pancreas that helps move glucose from your bloodstream into your body's cells—or with medications that increase your body's insulin production. If these treatments raise your insulin levels too high, your blood glucose can drop too low. Hypoglycemia can also occur if you do not eat when you need to or as much as you need, or if you skip a meal, drink too much alcohol, or exercise more than usual.

WHAT ARE THE SIGNS AND SYMPTOMS OF HYPOGLYCEMIA?

Hypoglycemia can be mild, moderate, or severe. If your blood glucose drops below normal (between 70 and 99 mg/dL), you may experience a variety of symptoms.

SIGNS AND SYMPTOMS OF HYPOGLYCEMIA	
Mild: below 70 mg/dL	• Hunger • Nervousness and shakiness • Sweating
Moderate: below 55 mg/dL	• Dizziness • Sleepiness • Confusion • Difficulty speaking • Feeling anxious or weak
Severe: below 35-40 mg/dL	• Seizure or convulsion • Loss of consciousness or coma

HOW IS HYPOGLYCEMIA TREATED?

If you have diabetes and you have symptoms of hypoglycemia, check your blood glucose level right away. If it's low, you should eat or drink something that will quickly raise your blood sugar. For mild to moderate hypoglycemia, you need to consume 15 grams of carbohydrates, such as glucose tablets, 5 or 6 pieces of hard candy, a cup of milk, 4 ounces of orange juice, or 6 ounces of regular (not diet) soda. Wait 15 minutes and retest your blood glucose level. If it is still low, consume another 15 grams of carbohydrates.

If hypoglycemia is not corrected right away, it can quickly worsen. You may become very confused and unable to manage your condition. In severe cases, you may even lose consciousness, have a seizure, or go into a coma.

Close friends and relatives should be aware of your condition and be taught how to recognize severe hypoglycemia and treat it quickly with an injection of glucagon (a hormone that raises blood glucose levels) if you cannot do it yourself. They will need to take you to the hospital or call for emergency assistance if

- Your consciousness is affected and no glucagon is available
- Confusion continues after treatment with glucagon
- Your blood glucose remains low despite eating carbohydrates or receiving glucagon

HOW CAN YOU AVOID HYPOGLYCEMIA?

The best way to avoid hypoglycemia is to eat meals and snacks on a regular schedule, test your blood glucose regularly, follow the exercise plan suggested by your diabetes health care team, and always take your diabetes medications as recommended. If needed, eat or drink something before and/or during exercise. Ask your doctor if your diabetes medication can cause hypoglycemia. If it can, ask whether you need to take additional precautions.

You can avoid hypoglycemia by being prepared. Be prepared to check your blood glucose level regularly and especially when you have any symptoms of hypoglycemia. Be prepared to treat those symptoms by keeping glucose tablets or hard candies handy. Make sure family and friends are prepared to help if you show signs of low blood glucose. Also, you should wear a diabetes identification bracelet or necklace to ensure proper emergency care if you are unable to speak for yourself.

You should check your blood glucose level before driving any vehicle or operating machinery to make sure it is in the normal range. This is especially important if you have repeated episodes of hypoglycemia or if you have trouble sensing when your blood glucose is low.

Questions to ask your doctor

- **Can my diabetes medicines cause hypoglycemia?**
- **If I get hypoglycemia, what should I do to bring my blood glucose level back to normal?**
- **Do I need a supply of glucagon?**
- **What should I tell my friends and relatives about hypoglycemia?**
- **Should I see a diabetes educator?**
- **Should I see an endocrinologist for my care?**

RESOURCES

- Find-an-Endocrinologist: *www.hormone.org* or call *1-800-HORMONE (1-800-467-6663)*
- Find a diabetes educator (American Association of Diabetes Educators): *www.diabeteseducator.org/DiabetesEducation/Find.html*
- Hormone Health Network information about diabetes: *www.hormone.org/diabetes*
- National Diabetes Information Clearinghouse (National Institutes of Health): *www.nlm.nih.gov/medlineplus/hypoglycemia.html#cat1*
- The American Diabetes Association: *www.diabetes.org*
- Mayo Clinic: *www.mayoclinic.com/health/hypoglycemia/DS00198*

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The Hormone Health Network offers free, online resources based on the most advanced clinical and scientific knowledge from The Endocrine Society (*www.endo-society.org*). The Network's goal is to move patients from educated to engaged, from informed to active partners in their health care. This fact sheet is also available in Spanish at *www.hormone.org/Spanish*.